Verification matrix

English version of ../../verification/matrix.md.

Check	What it proves	Command	CI
Backend tests	the domain and the API are correct	`make test-backend`	`ci.yml` → `backend`
Auth tests without PostgreSQL	the sign-in module is independent of the analytics	`make test-auth`	`ci.yml` → `backend`
ruff	static rules are respected (`E9`, `F` with an explicit config)	`make lint`	`ci.yml` → `backend`
Frontend dependency check	the UI builds reproducibly	`make test-frontend`	`ci.yml` → `frontend`
The API specification matches the code	the documentation does not lie about routes	`make openapi-check`	`portability.yml` → `contract`
Specification generation is idempotent	a repeated export does not change the file	`make openapi && git diff --exit-code backend/openapi.json`	`portability.yml` → `contract`
The stack is ready before the checks	the smoke test does not race the application	`make wait`	all `portability.yml` jobs
The showcase proxies the API	the nginx layer really works, not only the api	`make smoke-test` (checks `:8000` and `:8080`)	`portability.yml` → `compose-smoke`
Documentation is consistent	links are alive, variables and routes exist, every document has a translation and a PDF	`make docs-check`	`ci.yml` → `docs`, `portability.yml` → `docs`
Compose fails without a secret	the production configuration cannot be empty	`docker compose config`	`ci.yml` → `compose`
The full stack comes up	the services really talk to each other	`make compose-up` + `make smoke-test`	`portability.yml` → `compose-smoke`
A repeated start is safe	schema application is idempotent	two API starts in a row	`portability.yml` → `upgrade-path`
Backup → restore → smoke	data is restored, not only copied	`make backup` → `make restore` → `make smoke-test`	`portability.yml` → `backup-restore`
Start without optional integrations	AI, Telegram and e-mail are options, not a start condition	`make compose-up` without them	`portability.yml` → `config-without-secrets`
Host check	the environment is suitable before start-up	`make doctor`	locally

What is deliberately not checked

- LLM answer quality: only the fact that numbers come from the domain is checked.
- Performance and load (there is `backend/scripts/loadtest.py`, but it is not in CI).
- Several `api` replicas: Telegram session state and the snapshot live in the process.